

Python DSA Practice Sheet

Loops — Complete Concept Coverage | 50 Questions

□ Yeh sheet use kaise karein:

1. Pehle khud try karo bina solution dekhe. Google mat karo seedha.
2. Agar 15 minute mein solve nahi hua toh hint dekho — phir try karo.
3. Ek section khatam karo, tab agla section shuru karo.
4. Tricky section last mein karo — pehle foundation strong karo.
5. Roz minimum 5-6 questions karo — consistency > quantity.

SECTION 1: for Loop — Basic Iteration

□ `range()` ke teen forms yaad karo: `range(n)`, `range(a,b)`, `range(a,b,step)`. Yeh sab placement mein aate hain.

#	Question	Hint / Concept
1	1 se 10 tak ke saare numbers print karo.	<code>range(1, 11)</code>
2	1 se 100 tak ke sirf even numbers print karo.	<code>range(2, 101, 2)</code>
3	1 se 50 tak ke sirf odd numbers print karo.	<code>range(1, 51, 2)</code>
4	10 se 1 tak reverse order mein print karo.	<code>range(10, 0, -1)</code>
5	1 se n tak ke saare numbers ka sum nikalo. (n user se lo)	<code>sum += i</code>
6	1 se n tak ke saare numbers ka product (factorial) nikalo.	<code>prod *= i</code>
7	Ek number n ka table print karo (1 se 10 tak).	<code>n * i</code>
8	<code>range()</code> use karke 5, 10, 15, 20 ... 50 print karo.	<code>step=5</code>

SECTION 2: while Loop

□ `while` loop tab use hota hai jab hume pata nahi hota ki loop kitni baar chalega. Condition pe focus karo.

#	Question	Hint / Concept
9	<code>while</code> loop se 1 se 10 tak print karo.	<code>i < 11</code>
10	User se number lo aur <code>while</code> loop se uska reverse print karo (digits ek ek karke).	<code>n % 10</code> , <code>n //= 10</code>
11	Ek number tab tak 2 se divide karte raho jab tak 1 se bada rahe, steps print karo.	<code>while n > 1</code>
12	User se guess lo (1-10). Jab tak sahi guess na kare tab tak loop chale.	<code>while guess != target</code>
13	Digits ka sum nikalo (e.g. 1234 → 10) <code>while</code> loop se.	<code>n % 10</code>

SECTION 3: break aur continue

□ *break* = loop se bahar niklo. *continue* = current iteration skip karo. Dono placement mein commonly puchhe jaate hain.

#	Question	Hint / Concept
14	1 se 20 tak print karo, lekin 13 pe rok do (break use karo).	<i>if i == 13: break</i>
15	1 se 20 tak print karo lekin 3 ke multiples skip karo (continue use karo).	<i>if i % 3 == 0: continue</i>
16	List mein pehla negative number dhundo aur uske baad loop band karo.	<i>break on negative</i>
17	1 se 30 tak sirf prime numbers print karo (inner loop mein break use karo).	<i>flag + break</i>
18	User se numbers lo tab tak jab tak 0 na enter kare, sum print karo at the end.	<i>while True + break</i>

SECTION 4: Nested Loops

□ *Nested loops* = loop ke andar loop. *Patterns aur 2D problems* ke liye bahut zaroori hai. *Time complexity* $O(n^2)$ hoti hai.

#	Question	Hint / Concept
19	Right-angled star pattern print karo (5 rows): * ** *** **** *****	<i>2 for loops</i>
20	Inverted star pattern print karo (5 rows): ***** **** *** ** *	<i>range(n, 0, -1)</i>
21	Number pattern print karo: 1 1 2 1 2 3 1 2 3 4	<i>inner range(1, i+1)</i>
22	Multiplication table 1 se 5 tak (5x5 grid) print karo.	<i>nested for i, j</i>
23	Hollow square pattern print karo (N=5): sirf border mein * baaki spaces.	<i>if i==0 or i==n-1 or j==0 or j==n-1</i>
24	Pyramid pattern print karo (5 rows): * ** *** **** *****	<i>spaces + stars</i>

SECTION 5: Loops with Strings

□ *Python* mein string bhi iterable hai — har character pe loop chala sakte ho. *String problems* mein yeh concept bohot kaam aata hai.

#	Question	Hint / Concept
25	Ek string ke har character ko alag line pe print karo.	<i>for ch in string</i>
26	Ek string mein kitne vowels hain count karo.	<i>if ch in 'aeiouAEIOU'</i>

27	String ko reverse karo using loop (slicing mat use karo).	<i>reversed string build karo</i>
28	Check karo ki string palindrome hai ya nahi using loop.	<i>compare i aur n-i-1</i>
29	String ke saare uppercase letters lowercase mein convert karo without built-in lower().	<i>ord() aur chr()</i>
30	String mein consecutive duplicate characters remove karo. (e.g. 'aabbcc' → 'abc')	<i>compare current with previous</i>

SECTION 6: Loops with Lists

□ List traversal aur manipulation — DSA ka core concept. Index-based aur direct iteration dono practice karo.

#	Question	Hint / Concept
31	List ka maximum aur minimum element loop se dhundo (max()/min()) mat use karo.	<i>compare har element</i>
32	List ke saare elements ka average nikalo.	<i>sum / len</i>
33	List mein ek given element kitni baar aata hai count karo.	<i>counter variable</i>
34	List ke elements ko reverse order mein print karo without reverse().	<i>range(n-1, -1, -1)</i>
35	Do lists merge karo bina zip() ya + use kiye — loop se.	<i>index-based traversal</i>
36	List mein se saare duplicate elements remove karo, order maintain karo.	<i>seen set + loop</i>

SECTION 7: Number-Based Logic (Classic Problems)

□ Yeh questions almost har placement exam mein aate hain. Logic building ke liye best hain.

#	Question	Hint / Concept
37	N tak ke saare Fibonacci numbers print karo.	<i>a, b = 0, 1; swap</i>
38	Ek number prime hai ya nahi check karo using loop.	<i>divide 2 to sqrt(n)</i>
39	1 se N tak saare prime numbers print karo (Sieve concept samjho).	<i>nested loop + flag</i>
40	Ek number ka digit reversal karo using loop (e.g. 1234 → 4321).	<i>% 10 and // 10</i>
41	Armstrong number check karo (e.g. $153 = 1^3 + 5^3 + 3^3$).	<i>digit cube sum</i>
42	1 se 500 tak ke saare Armstrong numbers print karo.	<i>outer loop + inner digit logic</i>
43	GCD (HCF) of two numbers find karo using loop (Euclidean algorithm).	<i>while b != 0: a, b = b, a%b</i>
44	Ek number ke saare factors print karo.	<i>if n % i == 0</i>

SECTION 8: Tricky / Thinking Problems

□ Yeh thode mushkil hain — interview mein yahi puchhe jaate hain. Seedha code mat likho, pehle logic socho.

#	Question	Hint / Concept
45	Bina multiplication operator use kiye, loop se $A \times B$ nikalo.	<i>A baar B add karo</i>
46	Ek sorted list mein binary search implement karo using while loop.	<i>low, high, mid</i>
47	Bubble sort implement karo (nested loop se list sort karo).	<i>swap adjacent elements</i>
48	Loop se check karo ki koi number perfect hai ya nahi (sum of factors = number, e.g. 6).	<i>factor sum == n</i>
49	Two Sum: list mein do aise numbers dhundo jinki sum = target. (Brute force $O(n^2)$)	<i>nested loop</i>
50	Pascal's Triangle ke pehle N rows print karo using loops.	<i>prev row se next row banao</i>

Best of luck for your placements! □ — Made with Claude